

Maximising likelihood for multivariate Gaussian distributions

When you find the mean of some data, what you are really doing is finding a parameter which *maximises the likelihood*.

For instance, assume some points $\{\vec{x}_1, \dots, \vec{x}_N\} \subset \mathbb{R}^d$ are normally distributed. The conditional probability of the data is

$$P(\vec{x}_i | \vec{\mu}, \Sigma) = \prod_{i=1}^N \frac{1}{\sqrt{\tau^d \det \Sigma}} \exp\left(-\frac{1}{2}(\vec{x}_i - \vec{\mu})^T \Sigma^{-1}(\vec{x}_i - \vec{\mu})\right)$$

which is also the *likelihood* of the parameters $\vec{\mu}$ and Σ . It is often easier to manipulate the logarithm of the likelihood:

$$\log P = -\frac{1}{2} \log(\tau^d \det \Sigma) - \frac{1}{2} \sum_{i=1}^N (\vec{x}_i - \vec{\mu})^T \Sigma^{-1}(\vec{x}_i - \vec{\mu})$$

Fitting $\vec{\mu}$ to data

To find the mean $\vec{\mu}$ which maximises the likelihood, note that $\log P$ is quadratic in $\vec{\mu}$, so the maximum occurs at the unique point where its derivative vanishes.

Consider the differential $\delta \log P$ induced by $\vec{\mu} \rightarrow \vec{\mu} + \delta \vec{\mu}$:

$$\begin{aligned} \delta \log P &= \frac{1}{2} \sum_{i=1}^N \left[(\delta \vec{\mu})^T \Sigma^{-1}(\vec{x}_i - \vec{\mu}) + (\vec{x}_i - \vec{\mu})^T \Sigma^{-1} \delta \vec{\mu} \right] \\ &= \sum_{i=1}^N \left[(\vec{x}_i - \vec{\mu})^T \Sigma^{-1} \right] \delta \vec{\mu} \\ &= \left[\sum_{i=1}^N \vec{x}_i - N \vec{\mu} \right]^T \Sigma^{-1} \delta \vec{\mu} \end{aligned}$$

If the likelihood is at a local maximum, then $\delta \log P$ must vanish for any $\delta \vec{\mu}$. This holds when:

$$\vec{\mu} = \frac{1}{N} \sum_{i=1}^N \vec{x}_i \tag{1}$$

Fitting Σ to data

To find the covariance matrix Σ which maximises the likelihood, consider the differential likelihood induced by $\Sigma \rightarrow \Sigma + \delta \Sigma$.

$$\delta \log P = -\frac{N}{2} \frac{\delta \det \Sigma}{\det \Sigma} - \frac{1}{2} \sum_{i=1}^N (\vec{x}_i - \vec{\mu})^T \delta(\Sigma^{-1})(\vec{x}_i - \vec{\mu})$$

Use the identities

$$\begin{aligned} \delta \det A &= \text{tr}[A^{-1} \delta A] \det A \\ \delta(\Sigma^{-1}) &= \Sigma^{-2} \delta \Sigma = \delta \Sigma \Sigma^{-2} \end{aligned}$$

and take the trace to obtain

$$\text{tr}[\delta \log P] = -\frac{N}{2} \text{tr}[\Sigma^{-1} \delta \Sigma] - \frac{1}{2} \sum_{i=1}^N \text{tr}\left[(\vec{x}_i - \vec{\mu})(\vec{x}_i - \vec{\mu})^T \Sigma^{-2} \delta \Sigma\right]$$

where we use the cyclic property of the trace in the last term.

If the likelihood is at a local maximum, then it vanishes for any $\delta \Sigma$. Since $\delta \Sigma$ is arbitrary, this scalar equality between trace implies equality between the matrices themselves:

$$-2\delta \log P = -N \Sigma^{-1} \delta \Sigma + \sum_{i=1}^N (\vec{x}_i - \vec{\mu})(\vec{x}_i - \vec{\mu})^T \Sigma^{-2} \delta \Sigma$$

This vanishes when the covariance matrix is given by:

$$\Sigma = \frac{1}{N} \sum_{i=1}^N (\vec{x}_i - \vec{\mu})(\vec{x}_i - \vec{\mu})^T \tag{2}$$